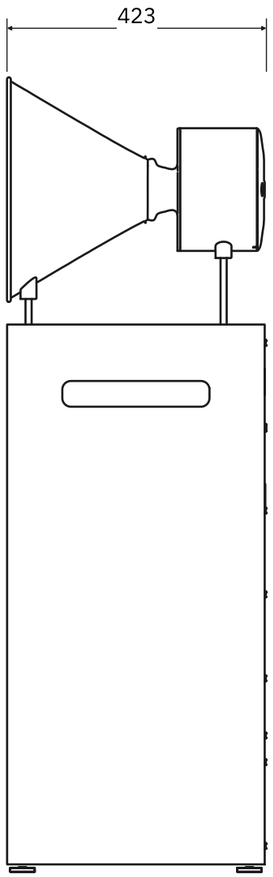
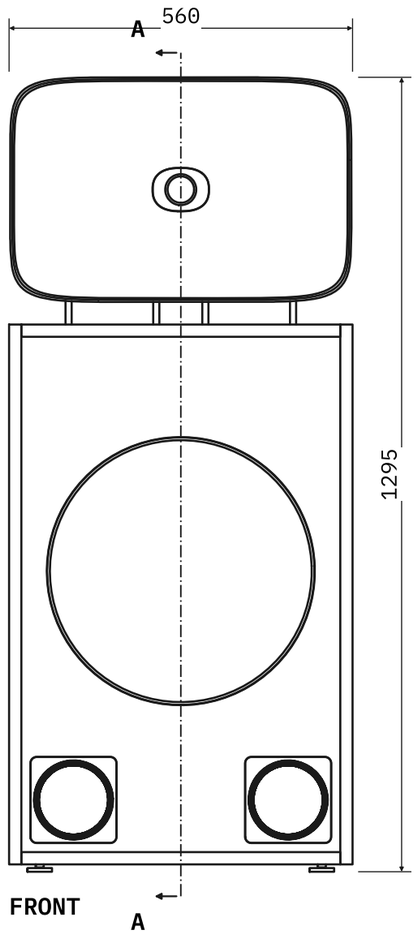
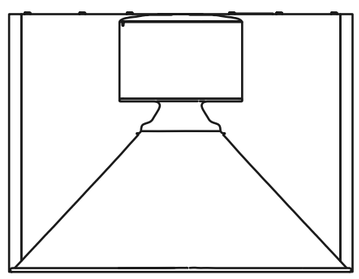
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

General arrangement

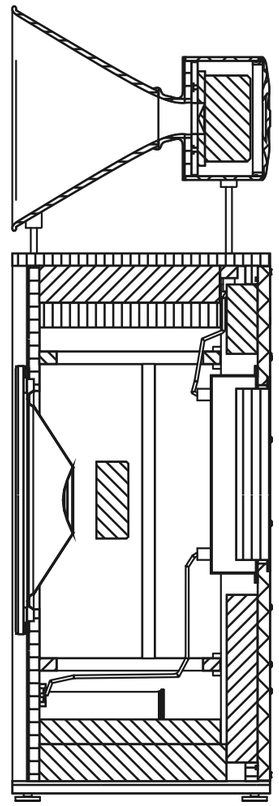
barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

BHC-001 rev A

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.0 and seen from the right; cut faces hatched, a different angle for each part.



Section A-A

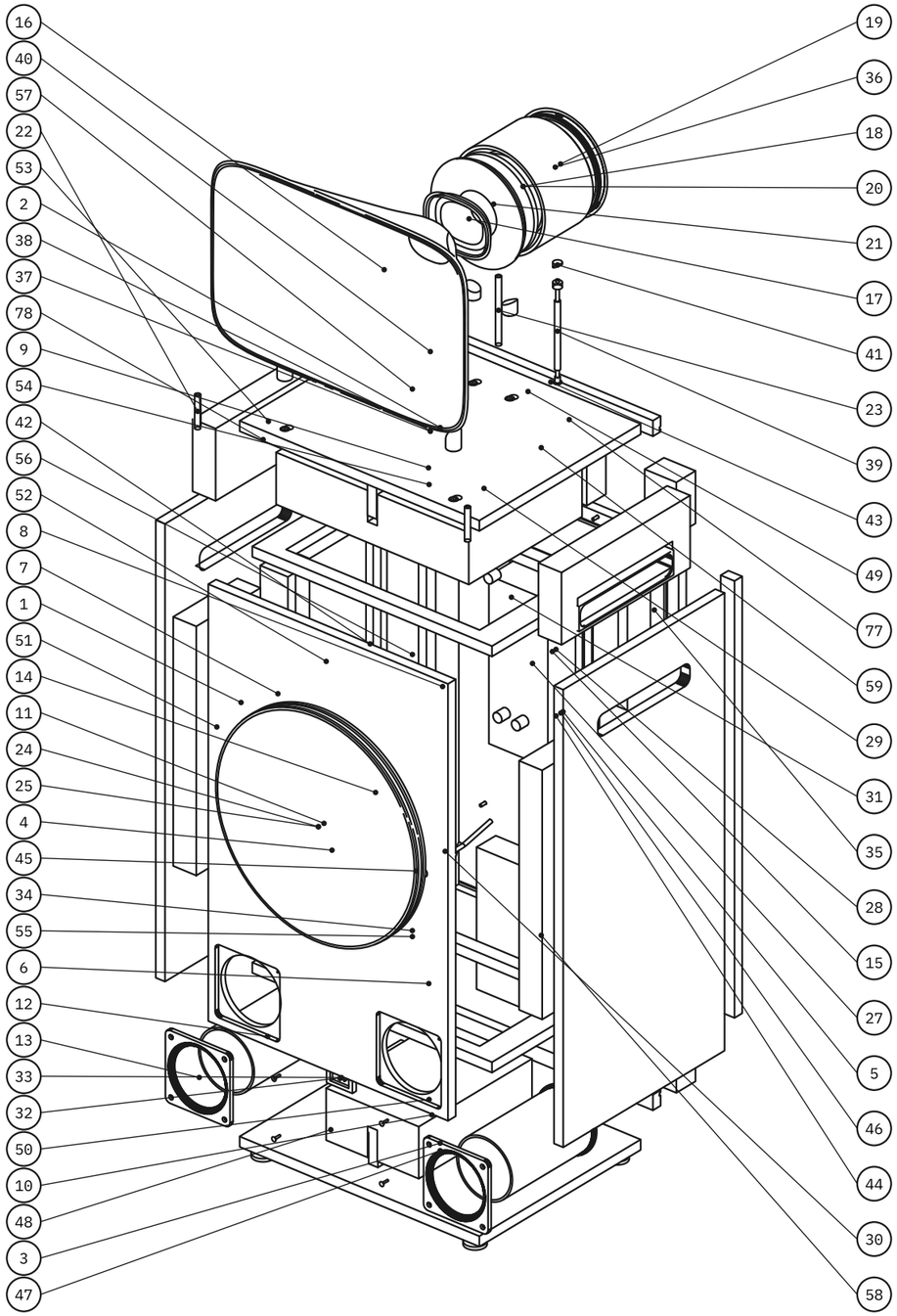
barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

BHC-002 rev A

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



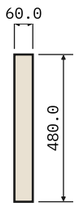
Exploded view

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

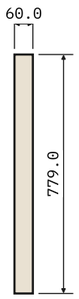
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:20. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



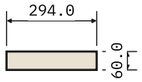
rib-left-1, rib-right-1

20 mm cast-acrylic, 4 for the set



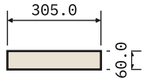
rib-back-1, rib-back-2

20 mm cast-acrylic, 4 for the set



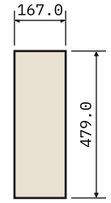
rib-top

20 mm cast-acrylic, 2 for the set



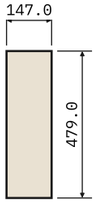
rib-bottom

20 mm cast-acrylic, 2 for the set



lining-left-2a, lining-right-2a

50 mm polyester-batting, 4 for the set



lining-left-2b, lining-right-2b

50 mm polyester-batting, 4 for the set

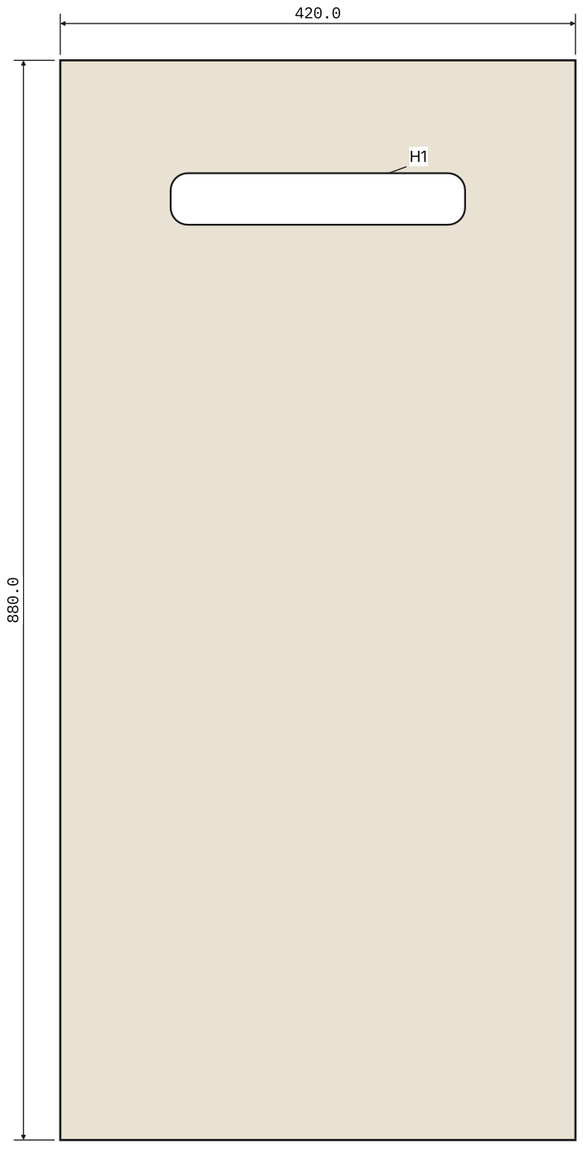
Sheet parts: plain panels

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/left.dxf. Marks listed at the right.



left: 20 mm cast-acrylic, 2 for the set, seen from its outer face

mark	size	x	y	depth
H1	profile (DXF)	210.0	767.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

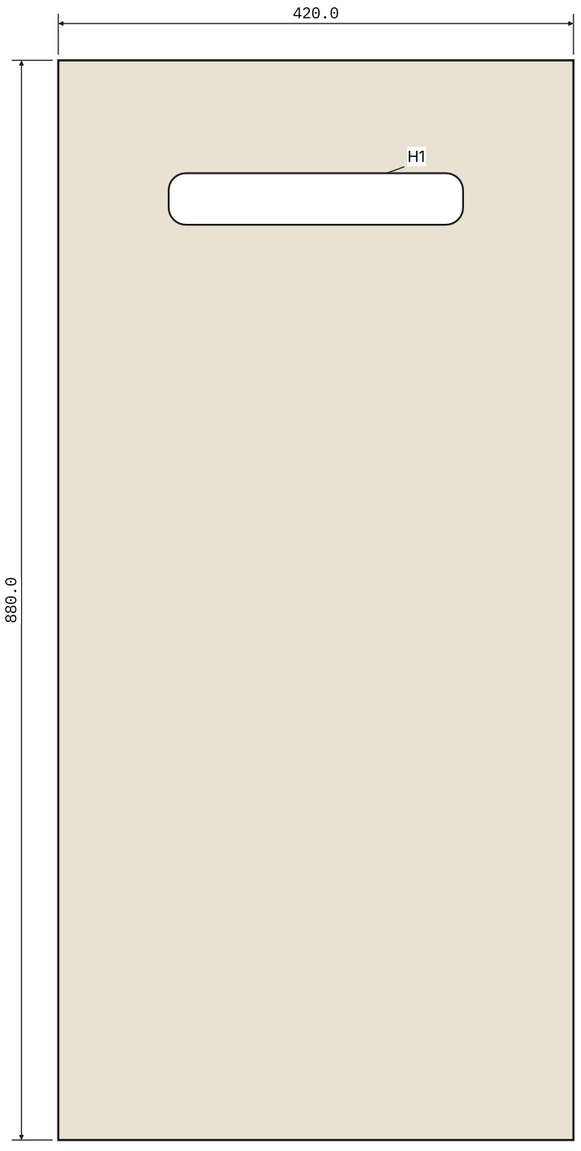
Sheet part: left

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/right.dxf. Marks listed at the right.



right: 20 mm cast-acrylic, 2 for the set, seen from its outer face

mark	size	x	y	depth
H1	profile (DXF)	210.0	767.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: right

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

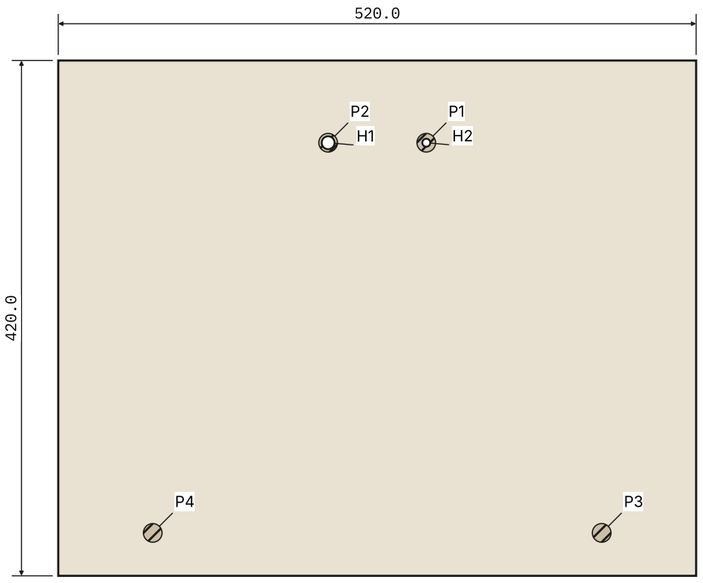
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/top.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø15.2	300.0	353.0	15 from face
P2	Ø15.2	220.0	353.0	15 from face
P3	Ø15.2	443.0	35.0	15 from face
P4	Ø15.2	77.0	35.0	15 from face
H1	Ø10.5	220.0	353.0	through
H2	Ø6.5	300.0	353.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



top: 20 mm cast-acrylic, 2 for the set, seen from its outer face

Sheet part: top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

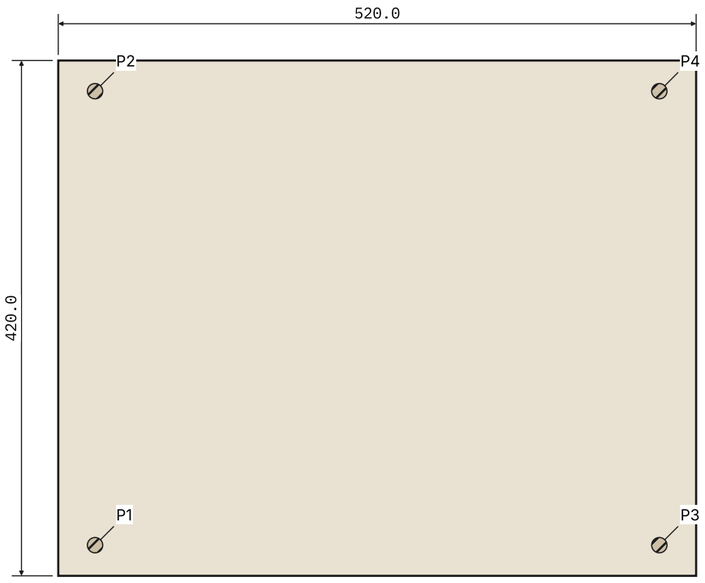
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/bottom.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø12.5	30.0	25.0	15 from face
P2	Ø12.5	30.0	395.0	15 from face
P3	Ø12.5	490.0	25.0	15 from face
P4	Ø12.5	490.0	395.0	15 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



bottom: 20 mm cast-acrylic, 2 for the set, seen from its outer face

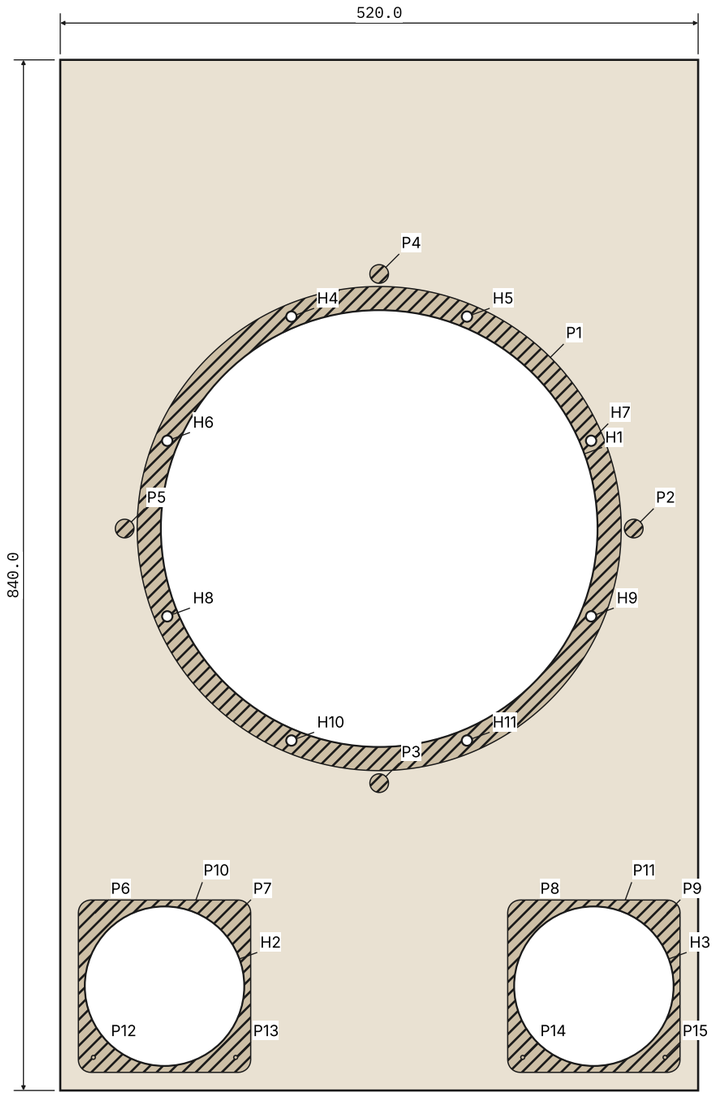
Sheet part: bottom

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/front-panel.dxf. Marks listed at the right.



front-panel: 20 mm cast-acrylic, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø394.6	260.0	458.0	11 from face
P2	Ø15.2	467.5	458.0	3.2 from face
P3	Ø15.2	260.0	250.5	3.2 from face
P4	Ø15.2	260.0	665.5	3.2 from face
P5	Ø15.2	52.5	458.0	3.2 from face
P6	profile (DXF)	27.0	143.0	17 from face
P7	profile (DXF)	143.0	143.0	17 from face
P8	profile (DXF)	377.0	143.0	17 from face
P9	profile (DXF)	493.0	143.0	17 from face
P10	profile (DXF)	85.0	85.0	9 from face
P11	profile (DXF)	435.0	85.0	9 from face
P12	profile (DXF)	27.0	27.0	17 from face
P13	profile (DXF)	143.0	27.0	17 from face
P14	profile (DXF)	377.0	27.0	17 from face
P15	profile (DXF)	493.0	27.0	17 from face
H1	Ø356.0	260.0	458.0	through
H2	Ø130.0	85.0	85.0	through
H3	Ø130.0	435.0	85.0	through
H4	Ø8.5	188.4	630.8	through
H5	Ø8.5	331.6	630.8	through
H6	Ø8.5	87.2	529.6	through
H7	Ø8.5	432.8	529.6	through
H8	Ø8.5	87.2	386.4	through
H9	Ø8.5	432.8	386.4	through
H10	Ø8.5	188.4	285.2	through
H11	Ø8.5	331.6	285.2	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

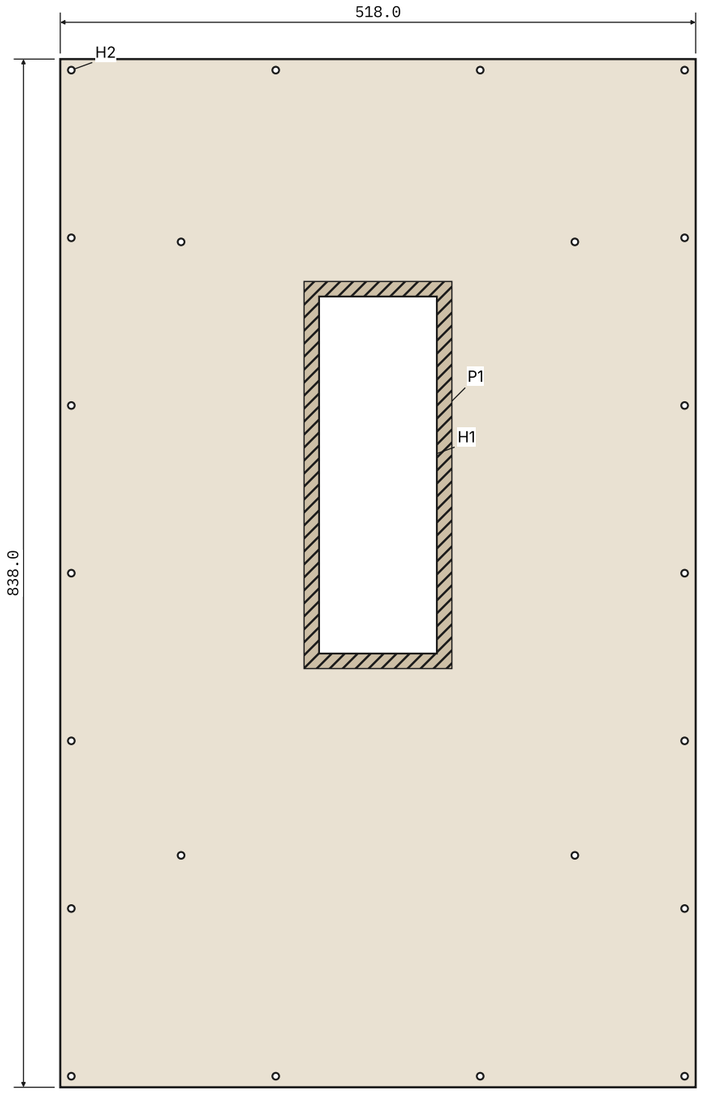
Sheet part: front-panel

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.



back: 20 mm cast-acrylic, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 × 315.5	259.0	499.0	4.5 from face
H1	96.0 x 291.0	259.0	499.0	through
H2	Ø5.5	22 off	DXF	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

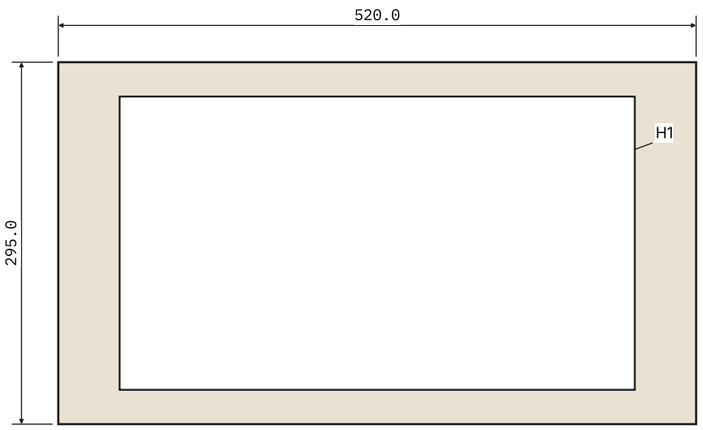
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 239.0	260.0	147.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace: 20 mm cast-acrylic, 2 for the set, seen from its outer face

Sheet part: brace

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

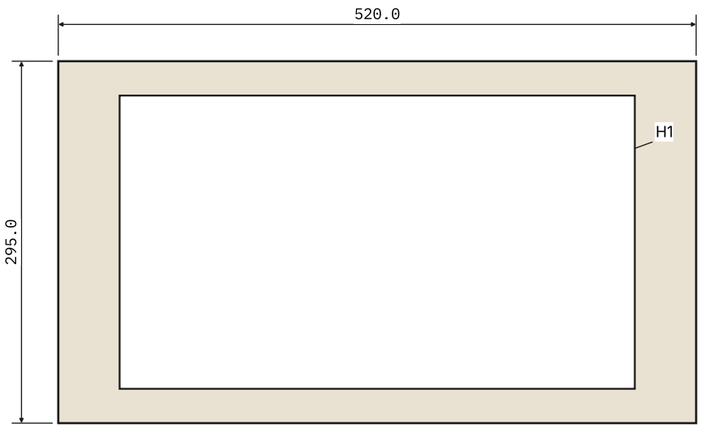
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace-2.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 239.0	260.0	147.5	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace-2: 20 mm cast-acrylic, 2 for the set, seen from its outer face

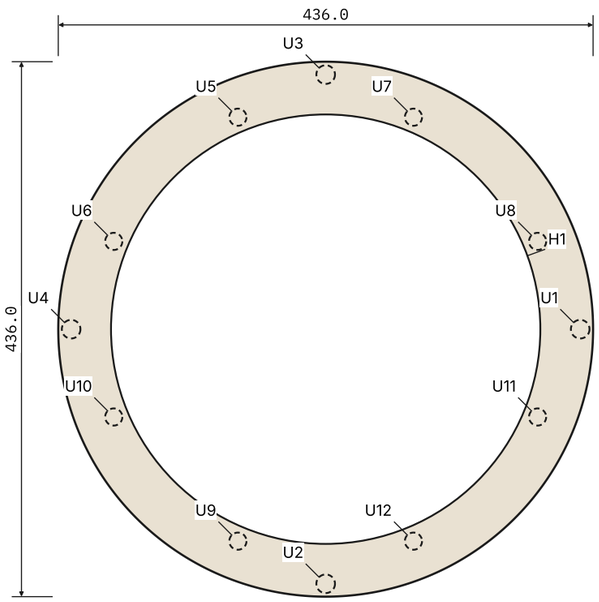
Sheet part: brace-2

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/woofer-ring.dxf. Marks listed at the right.



woofer-ring: 6 mm cast-acrylic, 2 for the set, seen from its outer face

inner-face pockets: dxf/woofer-ring-underside.dxf (seen from that face)

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	425.5	218.0	3.2 from inside
U2	Ø15.2	218.0	10.5	3.2 from inside
U3	Ø15.2	218.0	425.5	3.2 from inside
U4	Ø15.2	10.5	218.0	3.2 from inside
U5	Ø14.0	146.4	390.8	4 from inside
U6	Ø14.0	45.2	289.6	4 from inside
U7	Ø14.0	289.6	390.8	4 from inside
U8	Ø14.0	390.8	289.6	4 from inside
U9	Ø14.0	146.4	45.2	4 from inside
U10	Ø14.0	45.2	146.4	4 from inside
U11	Ø14.0	390.8	146.4	4 from inside
U12	Ø14.0	289.6	45.2	4 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

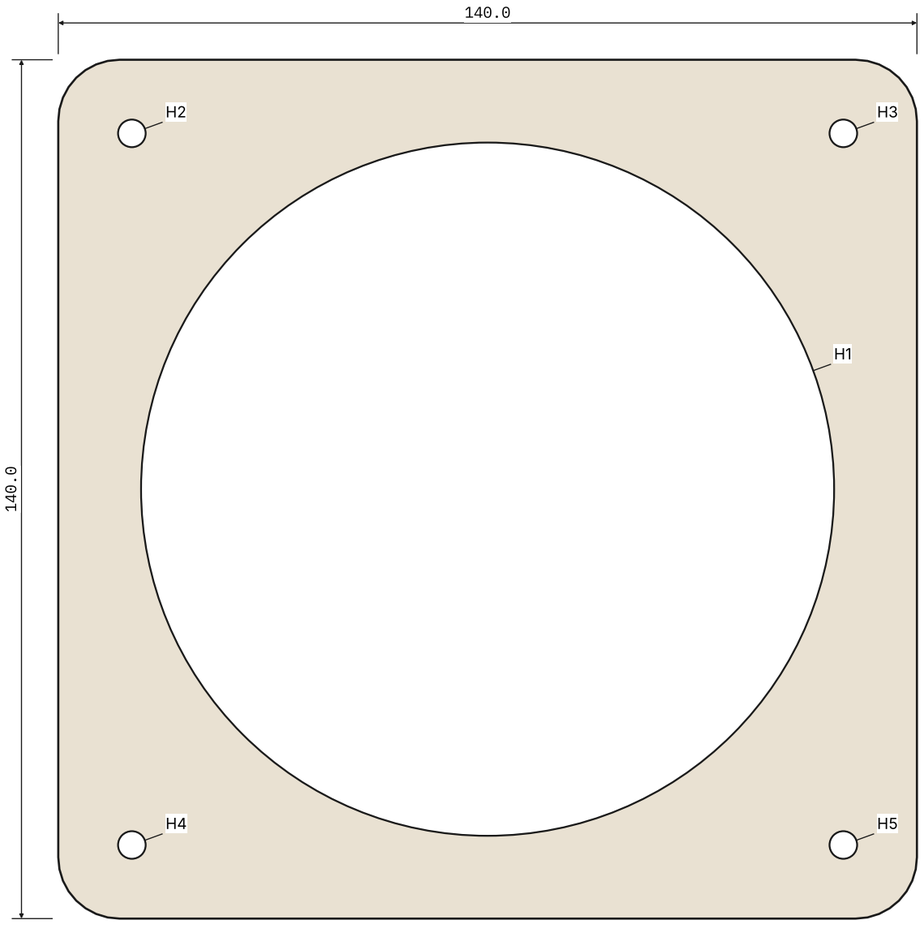
Sheet part: woofer-ring

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/port-frame-left.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	Ø113.0	70.0	70.0	through
H2	Ø4.5	12.0	128.0	through
H3	Ø4.5	128.0	128.0	through
H4	Ø4.5	12.0	12.0	through
H5	Ø4.5	128.0	12.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

port-frame-left: 8 mm cast-acrylic, 2 for the set, seen from its outer face

not in the cut file: 4 holes countersunk 90 deg from the outer face
and: the 113 hole at 70, 70 rounded over from the outer face (24 faces, 2 to 88 deg off the sheet)

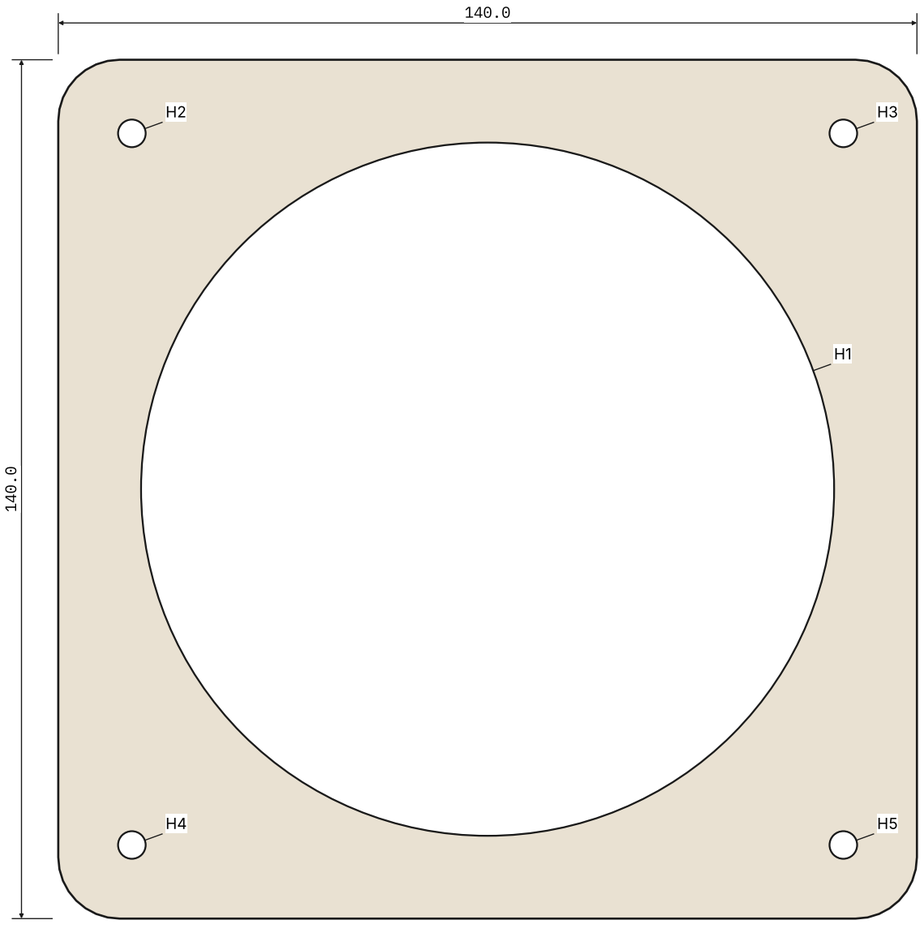
Sheet part: port-frame-left

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/port-frame-right.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	Ø113.0	70.0	70.0	through
H2	Ø4.5	12.0	128.0	through
H3	Ø4.5	128.0	128.0	through
H4	Ø4.5	12.0	12.0	through
H5	Ø4.5	128.0	12.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

port-frame-right: 8 mm cast-acrylic, 2 for the set, seen from its outer face

not in the cut file: 4 holes countersunk 90 deg from the outer face
and: the 113 hole at 70, 70 rounded over from the outer face (24 faces, 2 to 88 deg off the sheet)

Sheet part: port-frame-right

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

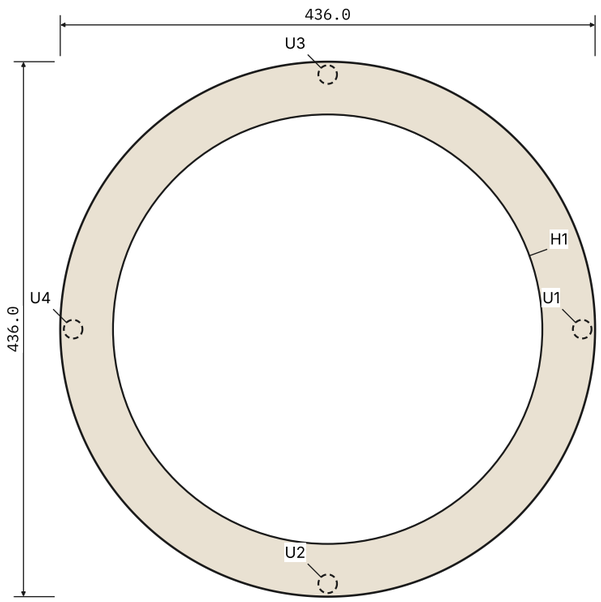
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/disc-frame.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	425.5	218.0	3.2 from inside
U2	Ø15.2	218.0	10.5	3.2 from inside
U3	Ø15.2	218.0	425.5	3.2 from inside
U4	Ø15.2	10.5	218.0	3.2 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



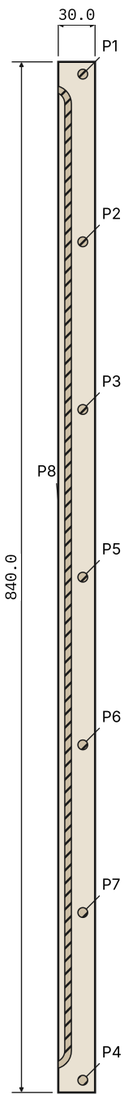
disc-frame: 12 mm cast-acrylic, 2 for the set, seen from its outer face
inner-face pockets: dxf/disc-frame-underside.dxf (seen from that face)

Sheet part: disc-frame

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/land-left.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø8.0	20.0	830.0	12 from face
P2	Ø8.0	20.0	693.3	12 from face
P3	Ø8.0	20.0	556.7	12 from face
P4	Ø8.0	20.0	10.0	12 from face
P5	Ø8.0	20.0	420.0	12 from face
P6	Ø8.0	20.0	283.3	12 from face
P7	Ø8.0	20.0	146.7	12 from face
P8	profile (DXF)	5.3	420.0	3 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

land-left: 20 mm cast-acrylic, 2 for the set, seen from its outer face

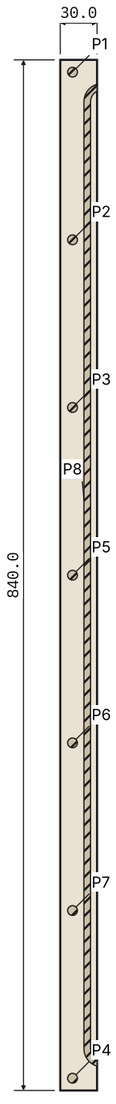
Sheet part: land-left

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/land-right.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	Ø8.0	10.0	830.0	12 from face
P2	Ø8.0	10.0	693.3	12 from face
P3	Ø8.0	10.0	556.7	12 from face
P4	Ø8.0	10.0	10.0	12 from face
P5	Ø8.0	10.0	420.0	12 from face
P6	Ø8.0	10.0	283.3	12 from face
P7	Ø8.0	10.0	146.7	12 from face
P8	profile (DXF)	24.6	420.0	3 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

land-right: 20 mm cast-acrylic, 2 for the set, seen from its outer face

Sheet part: land-right

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

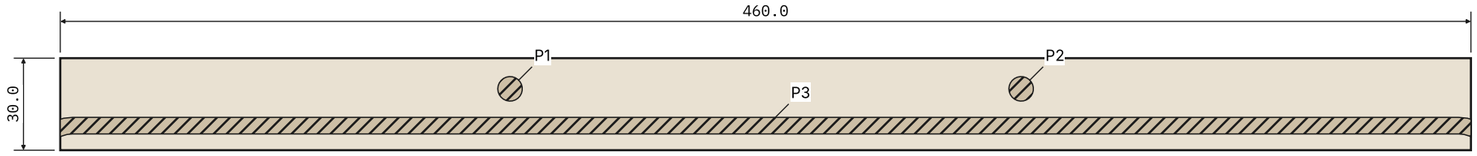
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/land-top.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø8.0	146.7	20.0	12 from face
P2	Ø8.0	313.3	20.0	12 from face
P3	profile (DXF)	230.0	7.5	3 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



land-top: 20 mm cast-acrylic, 2 for the set, seen from its outer face

Sheet part: land-top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

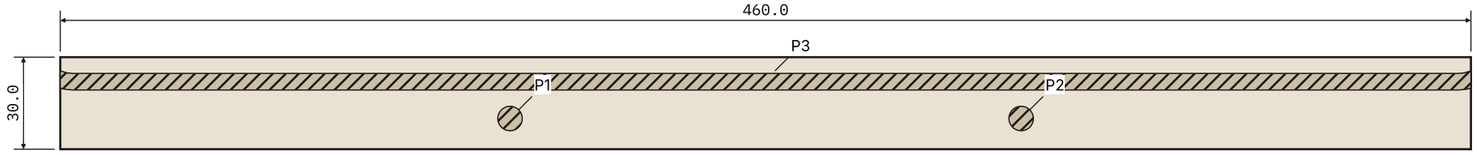
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/land-bottom.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø8.0	146.7	10.0	12 from face
P2	Ø8.0	313.3	10.0	12 from face
P3	profile (DXF)	230.0	22.5	3 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



land-bottom: 20 mm cast-acrylic, 2 for the set, seen from its outer face

Sheet part: land-bottom

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

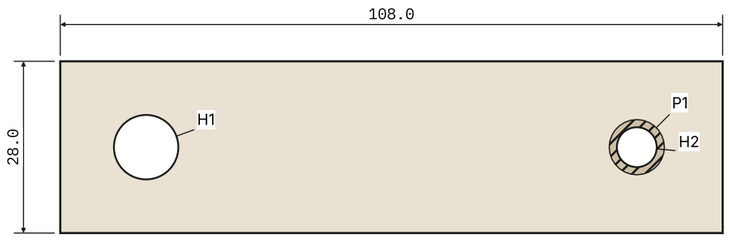
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

BHC-020 rev A

Scale 1:1. As the router sees it from the outer face; DXF: dxf/top-pad.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø9.0	94.0	14.0	10 from face
H1	Ø10.5	14.0	14.0	through
H2	Ø6.5	94.0	14.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



top-pad: 20 mm cast-acrylic, 2 for the set, seen from its outer face

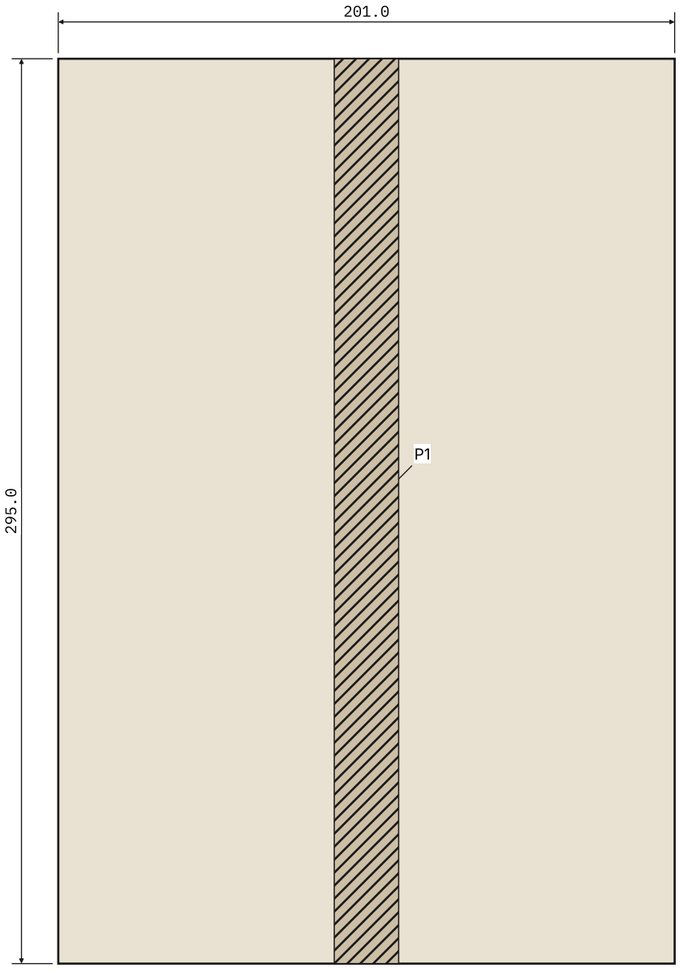
Sheet part: top-pad

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/lining-bottom-1.dxf. Marks listed at the right.



lining-bottom-1: 100 mm polyester-batting, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	21.0 x 295.0	100.5	147.5	60 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: lining-bottom-1

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

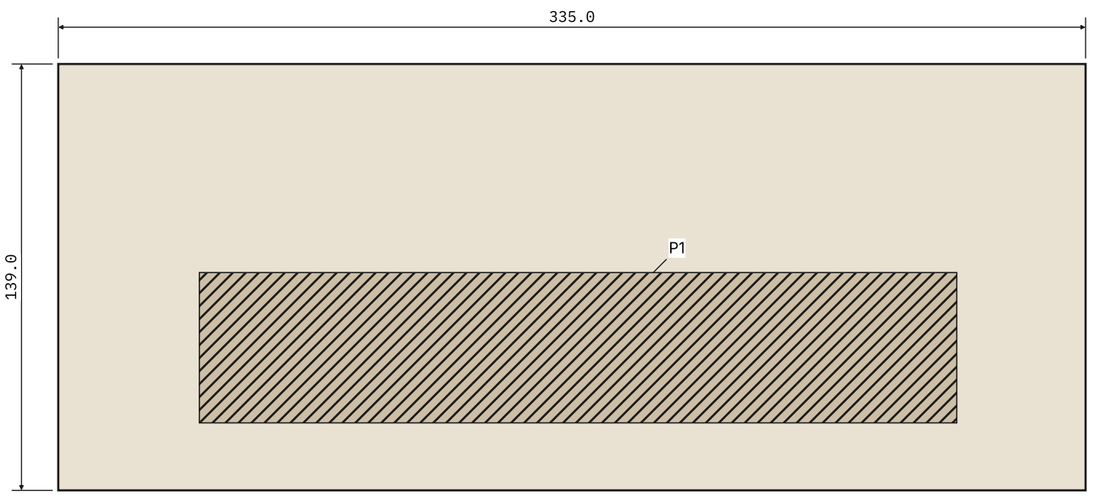
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/lining-left-3.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	247.0 x 49.0	169.5	46.5	43 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



lining-left-3: 50 mm polyester-batting, 2 for the set, seen from its outer face

Sheet part: lining-left-3

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

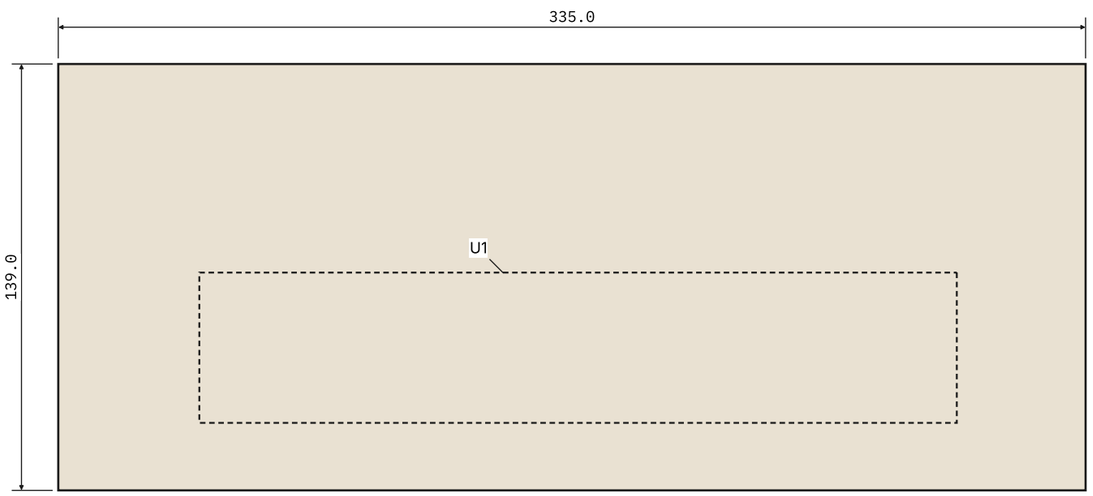
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/lining-right-3.dxf. Marks listed at the right.

mark	size	x	y	depth
U1	247.0 x 49.0	169.5	46.5	43 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



lining-right-3: 50 mm polyester-batting, 2 for the set, seen from its outer face

inner-face pockets: dxf/lining-right-3-underside.dxf (seen from that face)

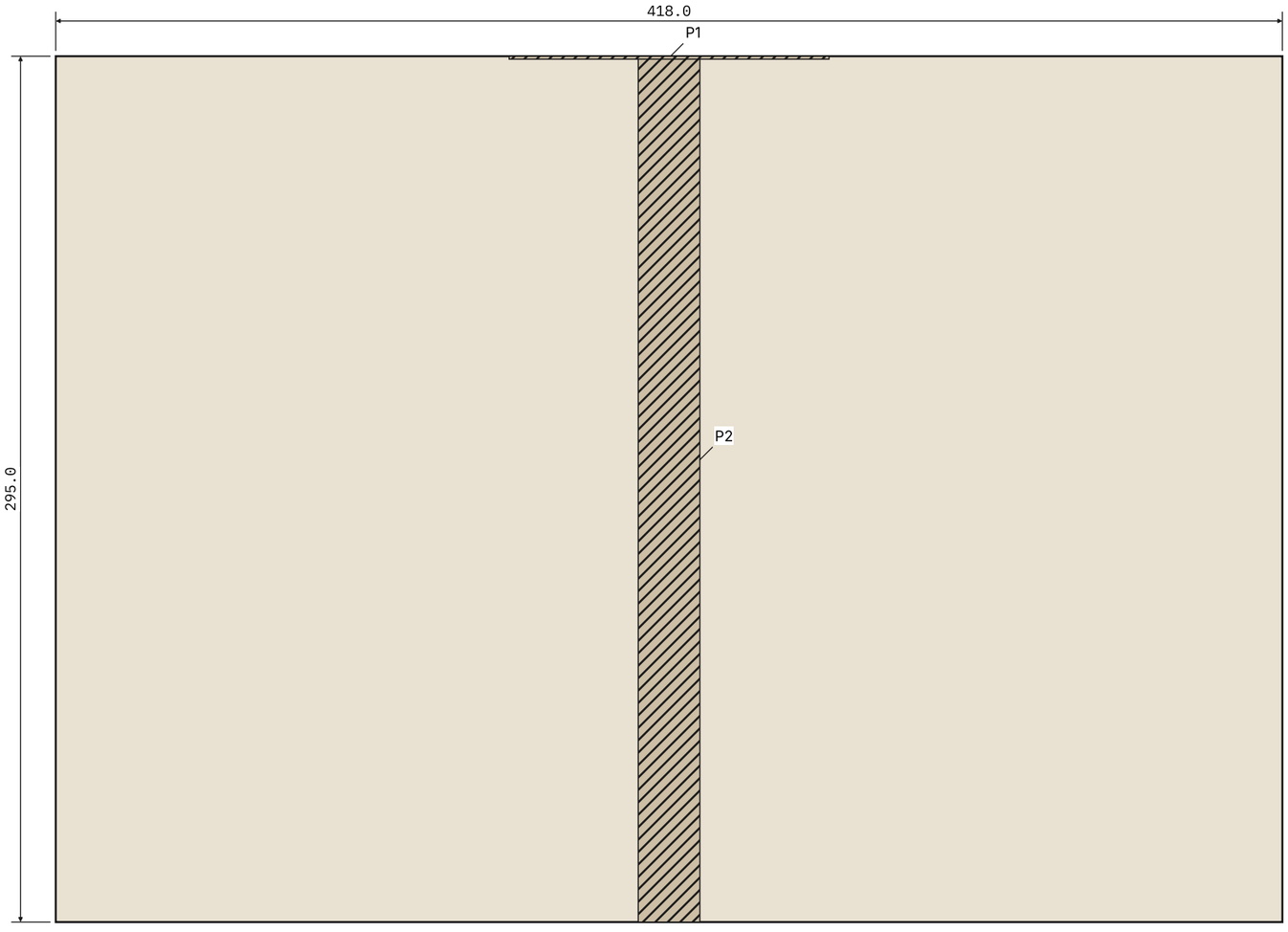
Sheet part: lining-right-3

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/lining-top-3.dxf. Marks listed at the right.



mark	size	x	y	depth
P1	109.0 x 1.0	209.0	294.5	20 from face
P2	21.0 x 294.0	209.0	147.0	60 from face

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

lining-top-3: 100 mm polyester-batting, 2 for the set, seen from its outer face

Sheet part: lining-top-3

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

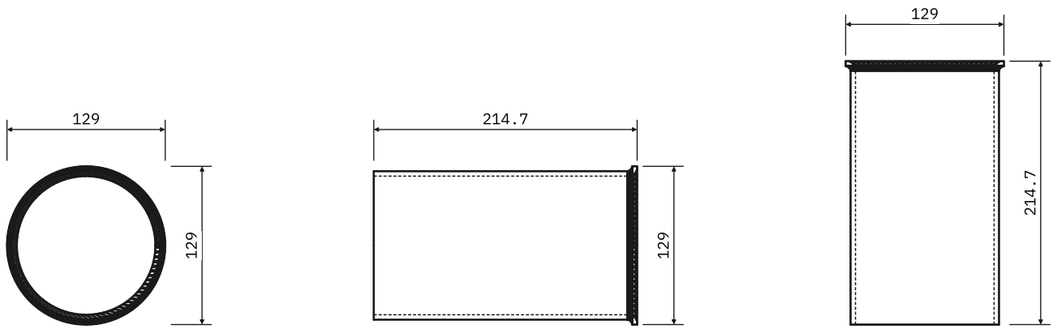
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

12. port-left, port-right: formed, cast-acrylic; 318 cm3 each; 4 for the set; 1:5

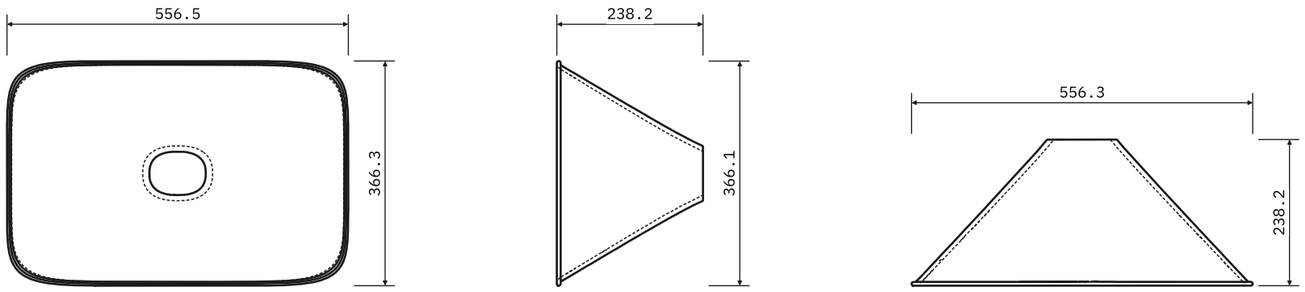
cast acrylic tube 121 × 4 (113 bore, polished), 215 long behind its frame, its inner end flared to R8 over a heated mandrel and annealed; cemented square into its frame's back face; the 2 together tune the box to 38 Hz; its length found on the first sample (three made 10 apart, the one whose impedance minimum falls at fb kept) and the rest made to it: its rounded ends cannot be trimmed



FRONT FROM THE RIGHT FROM ABOVE

16. waveguide: formed, cast-acrylic-thermoformed; 2008 cm3 each; 2 for the set; 1:10

mouth 536 × 346, 278 deep, 2 in throat, the air path as the simulation approved it; pressure-formed from 10 mm cast sheet tinted in the horn's own heavier dye lot (about 6 to 10 times the cabinet's dye, matched to a chip we approve, so its 5 to 8.5 mm wall reads deep and nearly opaque, as the site's pictures show the horn: the owner, 2026-10-31, "Deep, like the site") over a male aluminium tool (horn-tool): the blank dried, heated through at about 170 C, blown into a bubble, the tool raised into it and 4 bar over it; its face against the tool is the air path, exact; the outside free and ...



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

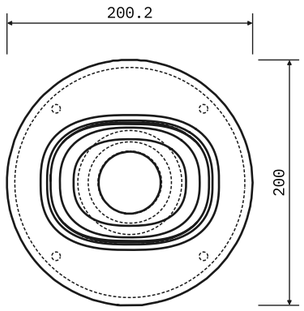
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

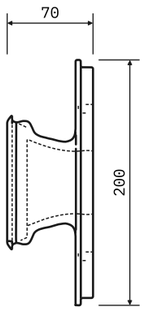
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

17. horn-throat: cnc-mill, cast-acrylic-horn-lot; 456 cm3 each; 2 for the set; 1:5

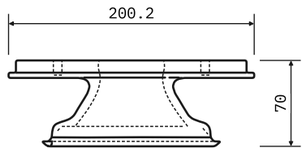
turned and milled from a 75 mm block of the horn's lot: its bore the air path from the driver's 51 exit to the trim, 7 of neck outside it rising out of the flange on an R10 cove, a 12 deep socket for the flare's end (its bore the flare's outside, 0.2 clear for the cement; the two air faces meet with no step, the throat's bore machined to the same surface as the tool), and a collar round it; its flange the driver's boss, 14 thick, its spigot in the pod; 4 M5 inserts bonded in blind holes on 170 take the adaptor ring



FRONT



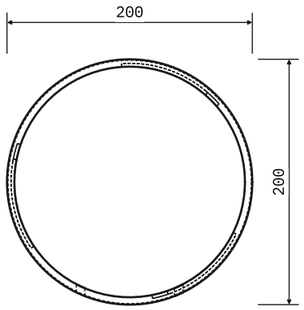
FROM THE RIGHT



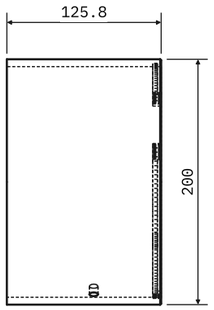
FROM ABOVE

18. horn-pod: formed, cast-acrylic-horn-lot; 456 cm3 each; 2 for the set; 1:5

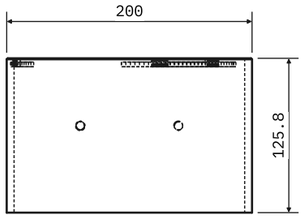
rolled from the horn lot's 6 mm sheet over a heated mandrel to 200 across, its seam scarfed and cemented underneath, between the rear posts; annealed, its bore turned true; cemented over the throat piece's spigot; the twist-lock's 3 L-slots milled in its bore; holed for the HF lead and the tie rod



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

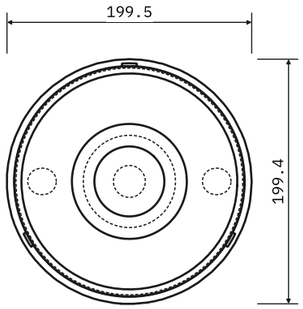
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

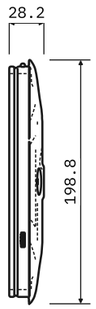
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

19. waveguide-cap: cnc-mill, cast-acrylic-horn-lot; 204 cm3 each; 2 for the set; 1:5

turned from the horn lot's 30 mm sheet, its three lugs left on the spigot and milled; closes the pod behind the driver on the twist-lock as before: an eighth of a turn (45 degrees, clockwise from behind), an O-ring on the spigot beyond the lugs; two shallow finger dimples to turn it by; no screw shows



FRONT



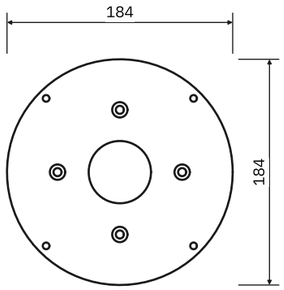
FROM THE RIGHT



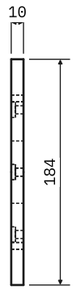
FROM ABOVE

21. hf-adaptor: lathe, aluminium; 242 cm3 each; 2 for the set; 1:5

184 across, 10 thick, bored 50.8 (the throat); 4 countersunk M6 on 101.6 take the driver from the front; 4 × 5.5 on 170 (at 45 degrees to them) take M5 socket caps into the brass inserts in the horn's boss; black anodised



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

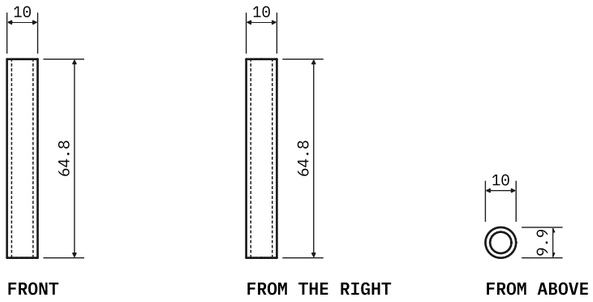
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

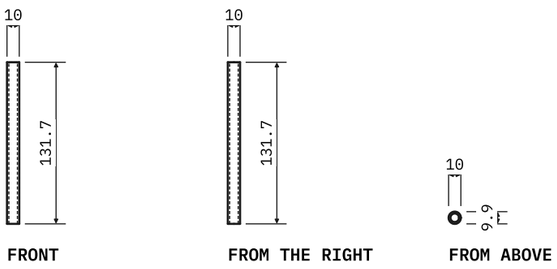
22. post-1, post-2: lathe, stainless; 3 cm3 each; 4 for the set; 1:2

65 long (42 showing): stainless tube 10 × 7, threaded M10 × 1 10 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred



23. post-3, post-4: lathe, stainless; 5 cm3 each; 4 for the set; 1:5

132 long (109 showing): stainless tube 10 × 7, threaded M10 × 1 10 at the top into the brass insert in its pad; its foot plain, standing 13 deep in a polyurethane sleeve in a brass socket in the top; its bore deburred; the HF lead runs down inside it



Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

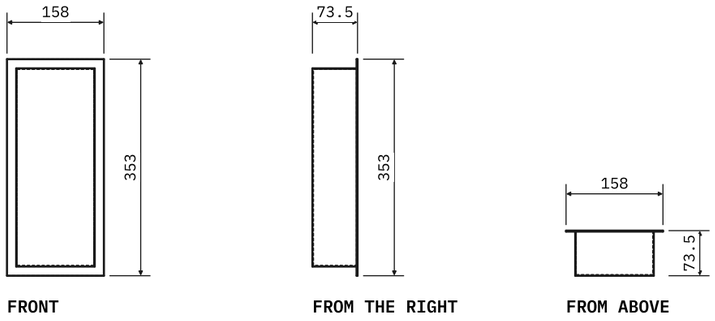
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

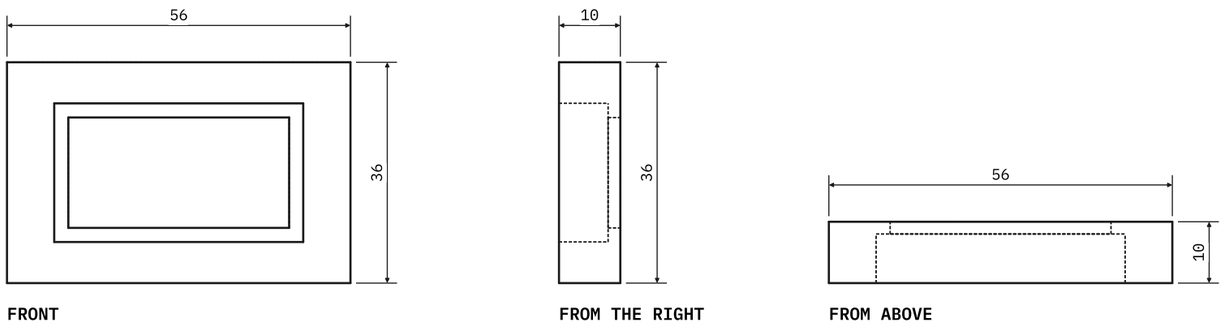
27. amp-box: sheet metal, aluminium-5052; 180 cm3 each; 2 for the set; 1:10

125 × 320 inside, 75 deep from the back: 1.5 aluminium 5052-H32 folded, its corners welded, black anodised; a 15 flange round its open side, on a 3 EPDM gasket on the back's inner face, ten screws through it, so the box stays airtight and the mains electronics are enclosed in metal; two M12 cable glands in its front; the ports' length found with it fitted (it takes volume)



33. ir-bracket: cnc-mill, clear-cast-acrylic; 12 cm3 each; 1.0 for the set; 1:1

56 × 36 × 10, milled from 10 mm clear cast acrylic, pocketed for the board, cemented to the front panel's back face; the board held in its pocket by two bonded M2.5 inserts and nylon screws (no screw into the panel)



Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

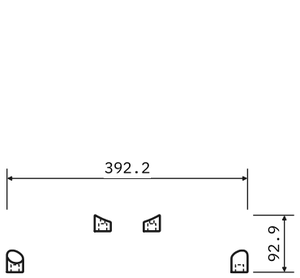
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

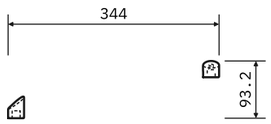
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

40. horn-pads: lathe, cast-acrylic-horn-lot; 40 cm3 each; 2 for the set; 1:10

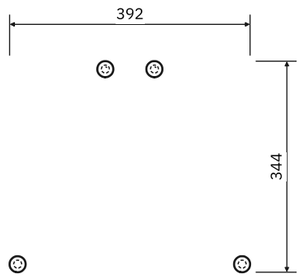
four pucks 26 across, turned from the horn lot's 30 mm sheet, their tops milled to the flare's (the rear pair's to the pod's) underside and cemented on through a jig taken off the cabinet's sockets; each foot 15 below the shell's lowest point over it, so the M10 × 1 insert (13 × 12, bonded) stays wholly in the pad; the rear pair bored 7 on through the pod for the HF lead and the tie rod



FRONT



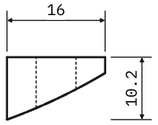
FROM THE RIGHT



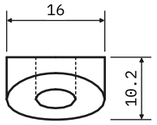
FROM ABOVE

41. horn-saddle: lathe, cast-acrylic-horn-lot; 1 cm3 each; 2 for the set; 1:1

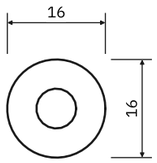
a puck 16 across cemented on the pod's floor inside at the tie post, its top flat for the tie rod's washer and thumb nut; bored 6.5



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

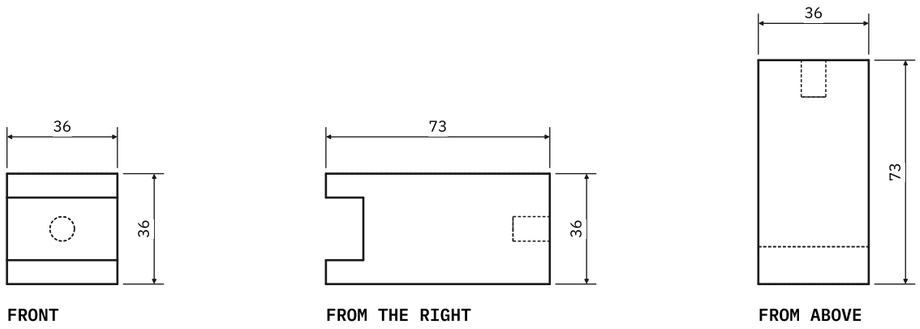
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

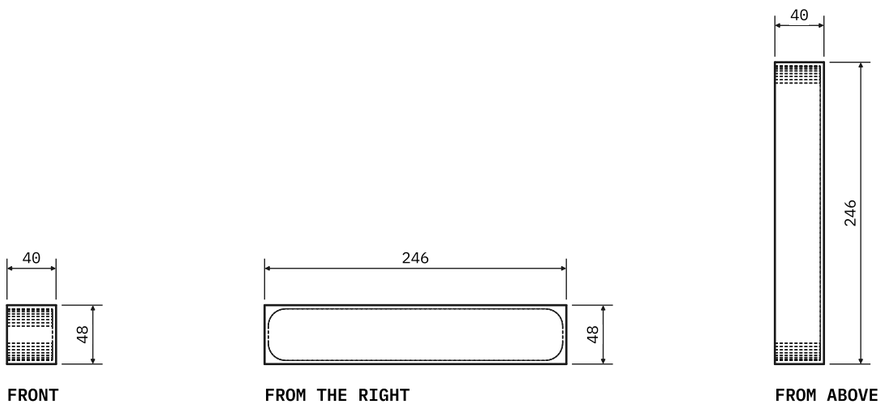
45. back-boss-1, back-boss-2, back-boss-3, back-boss-4: cnc-mill, cast-acrylic-block; 85 cm3 each; 8 for the set; 1:2

36 × 36, 73 deep: slotted 20 wide and 12 deep over the brace's rear member and cemented to it, its back face milled flush with the lands' so the back seats on both; an M5 insert bonded with epoxy in a 8 hole 12 deep takes the back's screw: the back tied to the brace



78. grip-left, grip-right: formed, cast-acrylic-formed; 106 cm3 each; 4 for the set; 1:5

a hand grip 240 × 42, its corners rounded on 14, cut through the side; a cup drape-formed from 3 mm acrylic in the side's tint over a buck, 40 deep, its flange trimmed and cemented behind the opening, so the box stays sealed



Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

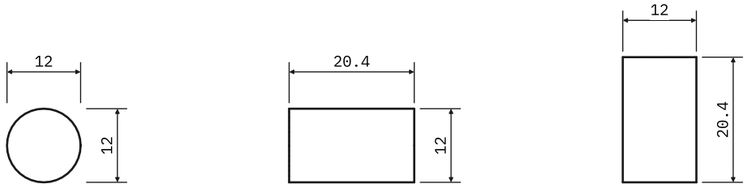
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

79. ir-window: lathe, clear-cast-acrylic; 2 cm3 each; 1.0 for the set; 1:1

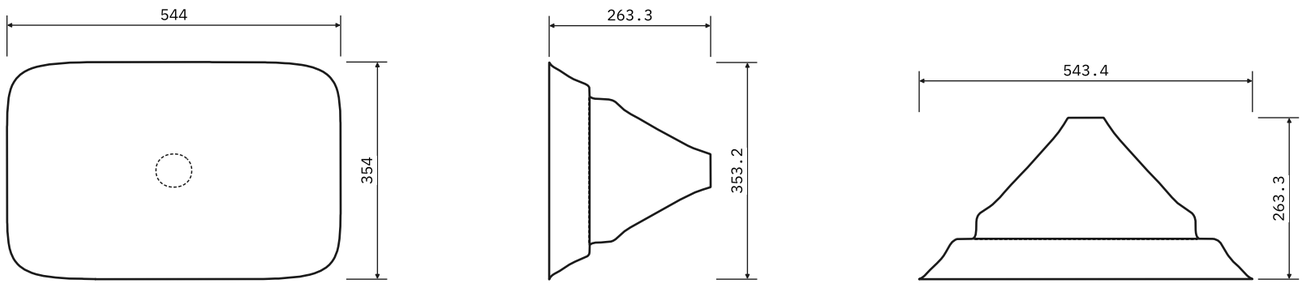
only if the 940 nm test fails: a clear cast acrylic rod 12 across, the panel bored through in front of the receiver's eye and the rod bonded in flush, its face polished; a clear dot in the tinted panel



FRONT FROM THE RIGHT FROM ABOVE

80. horn-tool-prototype: cnc-mill, tooling-board; 16725 cm3 each; 1.0 for the set; 1:10

the prototype's plug (the coordinator's call, round 36): the production plug's surface cut by the same 5-axis program from a glued-up block of high-temperature epoxy tooling board (rated 150 C or more), sealed and polished, vented as the aluminium; on the production base plate and trim fixture, which carry over. It forms the same 10 mm sheet at 4 bar over the same male geometry, so the prototype tests the production process, material and profile; its face is the air path, which sealed board holds as well as aluminium for a short run (some tens of shots), and the outside is the free ...



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

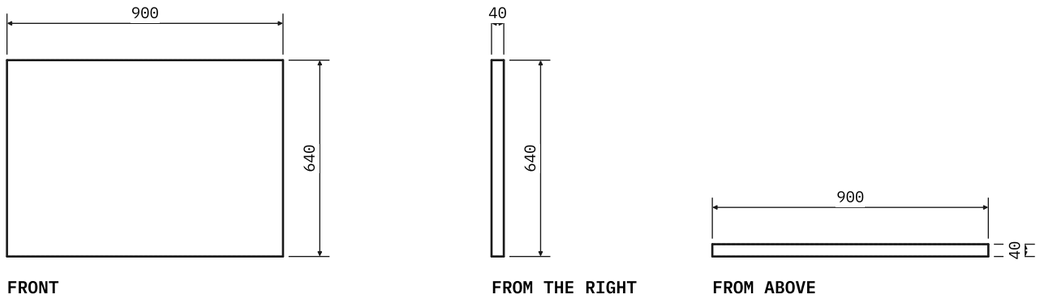
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

81. horn-tool-base: cnc-mill, aluminium; 23040 cm3 each; 1.0 for the set; 1:20

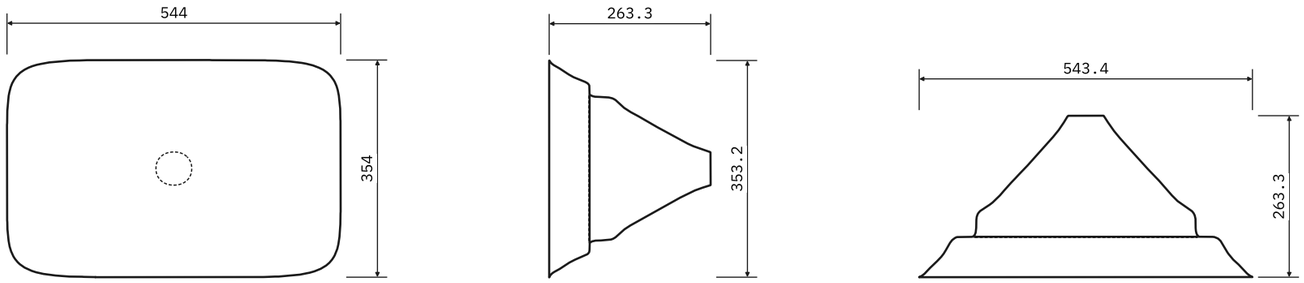
the tool's base plate in the mouth's plane, 6061 machined, the clamp frame's seal groove round it and the plug bolted on from below, so the flange forms flat on it; one tool makes every horn (the line's quantity is one)



FRONT FROM THE RIGHT FROM ABOVE

82. horn-tool: cast, aluminium; 16725 cm3 each; 1.0 for the set; 1:10

the forming tool's plug: the horn's air path from its nose 25 behind the trim (its edge rounded) to the mouth, its lip a cove R12 at its foot; A356 aluminium cast near shape from a pattern and machined all over on a 5-axis mill to the air path's surface, polished; drilled for oil at 85 C, vented by 0.6 mm holes in the waste and the mouth's corners; one tool makes every horn (the line's quantity is one)



FRONT FROM THE RIGHT FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	420 × 880	175.56
2	top	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	520 × 420	103.74
3	bottom	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	520 × 420	103.74
4	front-panel	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	520 × 840	207.48
5	back	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	518 × 838	206.19
6	brace, brace-2	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	520 × 295	72.87
7	rib-left-1, rib-right-1	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	60 × 480	13.68
8	rib-back-1, rib-back-2	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	60 × 779	22.20
9	rib-top	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	294 × 60	8.38
10	rib-bottom	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	305 × 60	8.69
11	woofer-ring	cut	2	6 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	436 × 436	27.09
12	port-left, port-right	formed	4	Cast acrylic tube, tinted through	318 cm3, 139 × 225 × 139	81.50
13	port-frame-left, port-frame-right	cut	4	8 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	140 × 140	3.72
14	woofer	buy	2	FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm)	393 frame, 356 cut-out, 165 deep	194.00
15	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 × 125 W into 4 ohm	315 × 120 × 55 plate, 291 × 96 recess	422.40
16	waveguide	formed	2	Cast acrylic sheet 10 mm, tinted through in the horn's own heavier dye lot per colour (about 6 to 10 times the cabinet's dye, Ember about 10; matched to a chip we approve; the lot's cost and minimum to confirm with the caster), pressure-	2008 cm3, 566 × 248 × 376	592.19
17	horn-throat	cnc-mill	2	Cast acrylic, tinted through in the horn's own heavier dye lot (about 6 to 10 times the cabinet's dye; 6, 30 and 75 mm), turned and milled	210 × 210 × 75	370.38
18	horn-pod	formed	2	Cast acrylic, tinted through in the horn's own heavier dye lot (about 6 to 10 times the cabinet's dye; 6, 30 and 75 mm), turned and milled	456 cm3, 210 × 136 × 210	155.10
19	waveguide-cap	cnc-mill	2	Cast acrylic, tinted through in the horn's own heavier dye lot (about 6 to 10 times the cabinet's dye; 6, 30 and 75 mm), turned and milled	210 × 210 × 35	157.18
20	hf	buy	2	B&C Speakers DE1085TN 2 in compression driver, titanium diaphragm, 100 mm voice coil, neodymium, 16 ohm (round 5: the owner's bigger throat)	154 across, 86 deep, 4 x M6 on 102	512.00
21	hf-adaptor	lathe	2	Aluminium 6061	194 × 20 × 194	43.82
22	post-1, post-2	lathe	4	Stainless steel 304	20 × 20 × 75	15.90
23	post-3, post-4	lathe	4	Stainless steel 304	20 × 20 × 142	16.70
24	disc-frame	cut	2	12 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	436 × 436	54.18
25	cloth-disc	buy	2	Acoustically transparent speaker cloth (a knit sold for grilles), about 0.6 × 0.95 m a grille, in the finish's colour		15.00
26	cloth-disc-magnets	buy	2	The ring's and the disc's magnets: eight 15 × 3 neodymium discs (N42) and four 15 x 3 mild-steel discs, a set: four magnets in the panel and the steel discs in the ring's back hold the ring, four in the disc's frame hold the disc to the same		10.00
27	amp-box	sheet metal	2	Aluminium 5052-H32 sheet, 1.5 mm, folded (6061 cracks at tight folds; the size column rounds the blank's 1.5 to 2; the engineering audit, 2026-11-01)	305 × 500 × 2	85.72

Parts list (1 of 4)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
28	amp-box-gasket	buy	2	EPDM sponge gasket, 3 mm, cut as a frame 15 wide for the amplifier compartment's flange		6.00
29	lead-hf	buy	2	HF lead: round silicone-sheathed cable 2 × 0.5 mm2, about 5 across, 1.3 m, ferrules at the pod's end and a two-way lever connector (Wago 221-412 or like) on the driver's own leads, a polarised 2-way connector at the compartment's end		6.00
30	lead-woofer	buy	2	Round-sheathed speaker cable 2 × 1.5 mm2, 1 m		2.00
31	cable-glands	buy	2	Three M12 nylon cable glands for 3 to 6.5 mm cable, black: two in the amplifier compartment's front, one under the top at the cable post, in an M12 insert for plastics (in acrylic, a thread tapped in the top)		4.00
32	remote-kit	buy	1.0	Hypex Fusion Remote Kit H08U003549: an IR receiver PCBA with a bi-colour LED, a 5-pin cable 125 cm to J15 on the FA122's input board, two light pipes, a tinted perspex window and screws; in the master speaker only (Hypex: "This set must be		22.00
33	ir-bracket	cnc-mill	1.0	Cast acrylic, water-clear (untinted)	66 × 20 × 46	37.43
34	lead-ir	buy	1.0	Hypex the Fusion Remote Kit's own 5-pin cable, 125 cm, to J15 on the FA122's input board (counted with the kit)		0.00
35	rating-plate	buy	2	Rating plate 70 × 40 × 1, black anodised aluminium, laser-marked: the model, the serial number, the ratings, the WEEE bin and the maker's address (round 35: the CE, UKCA and FCC marks only once the speaker has earned them); two M3 screws into		10.00
36	cap-o-ring	buy	2	O-ring for the horn's cap, 182 × 2 mm, EPDM 70 Shore A (round 16: the twist-lock), and a tube of silicone grease		4.00
37	post-sockets	buy	2	The posts' fittings (the engineering audit, 2026-11-01: make/engineering-audit.md): four M10 × 1 threaded inserts bonded with epoxy into the horn's pads (12 long), in holes drilled through a jig taken off the top's sockets; at the		90.00
38	post-sleeves	buy	2	Polyurethane sleeves 12 × 10 × 13, Shore A 90, a pair: pressed into the rear posts' sockets, they centre the posts and stop them buzzing (the front posts' feet sit in their keyhole sockets on PU washers)		8.00
39	tie-rod	buy	2	The horn's tie-down (the engineering audit, 2026-11-01: make/engineering-audit.md): an M6 A2-70 threaded rod about 110 long, a knurled M6 nut with a hex for a key and a washer, inside the pod, tightened to 2 N m (about 1.7 kN of		8.00
40	horn-pads	lathe	2	Cast acrylic, tinted through in the horn's own heavier dye lot (about 6 to 10 times the cabinet's dye; 6, 30 and 75 mm), turned and milled	30 × 30 × 110	49.95
41	horn-saddle	lathe	2	Cast acrylic, tinted through in the horn's own heavier dye lot (about 6 to 10 times the cabinet's dye; 6, 30 and 75 mm), turned and milled	26 × 26 × 20	15.68
42	land-left, land-right	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	30 × 840	11.97
43	land-top, land-bottom	cut	4	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	460 × 30	6.55
44	back-seal	buy	2	EPDM cord 4 mm, 60 Shore A, about 2.7 m, its ends vulcanised into a loop for the lands' groove (round 36: in place of the sponge gasket), and a spare		12.00
45	back-boss-1, back-boss-2, back-boss-3, back-boss-4	cnc-mill	8	Cast acrylic, tinted through, of the cabinet's own batch (sheet or block, 6 to 70 mm), turned and milled	46 × 83 × 46	18.78
46	back-screws	buy	2	M5 button heads (ISO 7380, A2 stainless, black: round 16) on black nylon washers, the back to the inserts in its land strips: 16 (M5 × 40 into HDPE, M5 × 30 into plywood); in the Cloud from round 36 22 x M5 × 30, 18 round the edge and 4 into		10.00

Parts list (2 of 4)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
47	feet	buy	2	Levelling feet M8, a 40 mm swivel base on a rubber pad, black, with lock nuts, a set of 4		18.00
48	port-screws	buy	2	The ports' fixings: eight black countersunk screws, four through each port's frame into the panel (thread-forming screws for plastics, EJOT PT K40 kind, 4 × 16 in the 20 mm HDPE; M4 × 16 into tapped threads in the 20 mm acrylic; wood screws 4 x		5.00
49	top-pad	cut	2	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	108 × 28	1.44
50	lining-bottom-1	cut	2	100 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	201 × 295	1.48
51	lining-left-2a, lining-right-2a	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	167 × 479	1.00
52	lining-left-2b, lining-right-2b	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	147 × 479	0.88
53	lining-left-3, lining-right-3	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	335 × 139	0.58
54	lining-top-3	cut	2	100 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	418 × 295	3.08
55	lining-back-a, lining-back-f	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	68 × 141	0.12
56	lining-back-b, lining-back-g	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	68 × 463	0.40
57	lining-back-c, lining-back-h	cut	4	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	68 × 101	0.09
58	lining-back-d	cut	2	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	239 × 273	0.82
59	lining-back-e	cut	2	50 mm Polyester batt, acoustic grade, about 30 kg/m3, thermally bonded (no binder), with recycled PET content	239 × 113	0.34
60	woofer-fixings	buy	2	M6 brass threaded inserts, knurled, bonded with epoxy into 8.5 holes in the acrylic panel, with M6 × 25 button heads (A2-70) tightened to 5 N m, not 15: an A2-70 M6 yields near 9 N m, and the bonded insert is the weaker part (the		14.00
61	hf-fixings	buy	2	The 2 in driver's fixings: four M6 × 16 countersunk screws (A2), the driver to its adaptor ring, and four M5 × 16 socket caps with washers, the ring to the brass inserts in the horn's boss, a set		6.00
62	cement	buy	2	Two-part polymerising acrylic cement for thick cast sheet (ACRIFIX 2R 0190 or Weld-On 40; round 35: not 1R 0192, a one-part cement that cures by light and will not cure through the deep tints), clear, about 0.5 kg a speaker: the corners, the		45.00
63	gasket	buy	4	1.5 mm closed-cell foam gasket strip, 1 m		5.00
64	lining-sleeves	buy	2	White polyester voile (a sheer curtain fabric), about 3 m2 a speaker with its seams, sewn into a sleeve round each batt of the lining, so no fibre reaches the ports or the woofer and the batt's face stays even		10.00
65	lining-tape	buy	2	Clear acrylic foam tape (3M VHB 4910 or alike), 20 mm dots, about 60 a speaker: on a Cloud cabinet four hold each batt's sleeve flat on the acrylic, all but invisible through it		12.00

Parts list (3 of 4)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
66	amp-fixings	buy	2	The FA122's plate in acrylic: ten M4 brass inserts bonded with epoxy into the back on the plate's ten-hole pattern (five each side; the model's holes were measured off Hypex's drawing, to be confirmed against it at first article (the engineering		16.00
67	earth-bond	buy	2	The amplifier compartment's earth bond (round 35, the engineering review's B3): an M4 stainless stud and nut, a star washer, and a green and yellow lead 1.0 mm2 about 150 long with M4 ring terminals, from a bared spot on the compartment to the		4.00
68	amp-box-fixings	buy	2	Ten screws for the amplifier compartment's flange, with washers: thread-forming screws for plastics (EJOT PT K40 × 12 kind) into HDPE, M4 screws into inserts bonded in acrylic, or wood screws into plywood		3.00
69	remote	buy	1.0	Hypex Hypex Remote Control H08U001537, IR (Philips RC-5, device 16: power 50, volume up 51, down 52, mute 53), its CR2032 included; one a pair, not supplied with the amplifier		11.00
70	link-cable	buy	1.0	75 ohm coaxial cable with RCA plugs, 5 m (Hypex: under 25 m), the master FA122's S/PDIF I/O to the slave's: Hypex's link carries the volume and the remote's commands to the second speaker, whose own IR is disabled		15.00
71	hf-capacitor	buy	2	33 uF 100 V polypropylene film capacitor, in series with the driver (a first-order high-pass at about 300 Hz into 16 ohm), clipped on nylon standoffs 3 mm off the compartment's pan (the engineering audit, 2026-11-01: the L-pad is gone; the HF		12.00
72	mains-cord	buy	2	Mains cord set, NEMA 5-15P to IEC 60320 C13, 2 m, 18 AWG SJT, black, UL listed (UL 817) (the engineering audit, 2026-11-01: make/engineering-audit.md): a listed cord is part of the end product's listing		9.00
73	loom	buy	2	The loom: cable-tie bases every 120, with thread-forming screws for plastics (EJOT PT K30 × 10 kind) in HDPE and on bonded M3 brass inserts with M3 screws in acrylic (round 35: nothing self-tapping into acrylic, which cracks round a formed thread),		18.00
74	terminals	buy	8	6.3 mm insulated push-on terminal		0.50
75	back-inserts	buy	2	M5 brass threaded inserts, knurled, a set of 22 (16 before round 36), bonded with epoxy into 8 holes 12 deep in the back's acrylic land strips and the braces' four bosses		13.00
76	feet-inserts	buy	2	M8 brass threaded inserts, knurled, a set of 4, bonded with epoxy into 12.5 holes 15 deep in the acrylic bottom, for the feet		5.00
77	mark-back	buy	2	Laser etching into the cast acrylic, frosted, about 0.1 deep, from a vector file: as standard (round 24) EARF 140 wide high on the back, the etch filled with the speaker's own colour (a deeper tint of it, wiped in and cured)		25.00
78	grip-left, grip-right	formed	4	Cast acrylic sheet 3 mm, tinted through, drape-formed over a buck	106 cm3, 50 × 256 × 58	39.65
79	ir-window	lathe	1.0	[option: IR window, fallback] Cast acrylic, water-clear (untinted)	22 × 30 × 22	25.59
80	horn-tool-prototype	cnc-mill	1.0	[option: tooling, the prototype's] High-temperature epoxy tooling board, glued up as a block, CNC-cut, sealed (the prototype's forming plug; lead time 3 to 4 weeks from the program)	554 × 273 × 364	4,180.67
81	horn-tool-base	cnc-mill	1.0	[option: tooling, the prototype's] Aluminium 6061	910 × 50 × 650	2,744.38
82	horn-tool	cast	1.0	[option: tooling, production (after the first pair)] Aluminium 6061	16725 cm3, 554 × 273 × 364	14,464.93

Parts list (4 of 4)

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. The waveguide stands on four posts, stainless lamp tube 10 × 7, each threaded M10 × 1 at its top into a brass insert bonded in a pad cast under the horn (drilled after casting, through a jig taken off the top's sockets), the rear posts' plain feet standing in polyurethane sleeves in brass sockets bonded into the top, and the front posts' feet, turned heads 14 across, hooked under the lips of keyhole sockets bonded through the top into the front panel's edge (the engineering audit, 2026-11-01: loose feet let the horn lift at a push of about 13 to 23 N), so the horn and its posts lift off as one once slid 10 mm forward. An M6 rod down the free rear post, from a knurled nut inside the pod tightened to 2 N m to an insert under the top, holds the rear down and locks the slide: held at three points, the horn cannot lift. The cap closing the pod is held by a twist-lock (round 16): three lugs on its spigot in L-slots in the pod's bore, locked by an eighth of a turn, an O-ring beyond them; no screw shows. The HF lead, a round silicone cable 5 across, runs from its connector at the amplifier's compartment straight up, through a gland under the top and up inside a rear post into the pod, where it plugs into the driver's leads. To lift the horn off: the cap off, the nut undone, the lead unplugged, and the horn slid 10 mm forward.
2. The FA122 crosses at 1 kHz, LR4 (the one crossover in every file since the engineering audit of 2026-11-01; round 35: in its simulation the horn loads the driver fully only from about 900 Hz, and two-way speakers of this size cross where their horns allow: JBL's 4367 at 700 Hz and its M2 at 800 on deeper waveguides, its 4430 at 1 kHz; 800 to 900 Hz only if the first sample's distortion at 100 dB/1 m stays low; B&C asks 12 dB/oct or steeper, at 800 Hz or above) with the HF channel about 6 dB down (105 dB at 2.83 V against the woofer's 99), all of it in the FA122's DSP (the engineering audit, 2026-11-01: no L-pad; the bare driver's hiss is about 8 to 10 dB SPL at 1 m, inaudible, and the FA122's full 22.4 V is 31 W into the driver, within its 140 W rating), behind a 33 uF film capacitor that keeps low frequencies off the driver if the preset is ever lost.
3. The amplifier's preset, stored before the speaker ships, the same verified preset in all three of the FA122's slots so no slot plays without it, behind Hypex's filter password set at the factory and recorded in the first-article file (the engineering audit, 2026-11-01): the crossover; a 4th-order Butterworth high-pass at 28 Hz on the woofer (below the ports' tuning the box no longer holds the cone, which would pass Xmax below about 30 Hz at full output, and a record's warp drives it there); limiters set to the woofer's Xmax and the driver's rating; soft clip on; a notch at the ports' pipe resonance (about 540 Hz in the pressed cabinet, 560 in the Cloud's, 600 in the plywood one), measured in each port's near field; and Hypex's auto-standby on (round 35: about 0.5 W asleep, and the compartment cooler in a warm room).
4. The amplifier in a sealed compartment of its own, as Hypex asks: 1.5 mm aluminium 5052-H32 (folds without cracking, where 6061 would not), black anodised, on an EPDM gasket on the back's inner face, so the box stays airtight and the mains electronics are enclosed in metal. The leads leave it through cable glands. A rating plate beside the amplifier carries the model, the serial number, the ratings, the marks and the maker's address; the finished speaker needs an EMC pre-scan and an IEC 62368-1 assessment.
5. The lining: polyester batts (thermally bonded, recycled PET), 50 mm on the sides and 100 mm on the top and the bottom (round 21), one to each bay between the braces (and the ribs), and 50 mm on the back, its full height between its ribs, so they come off with it; each cut 10 full each way and sewn into a sleeve of white voile; none on the front panel, none beside the ports (10 clear of each tube, nothing between its inner end and the back's batts) and none within 20 of the amplifier's compartment.
6. The back comes off (round 14): it closes on a 3 mm EPDM gasket against four land strips inside the shell, sixteen M5 button heads through it into inserts in them. The amplifier's compartment, the back's ribs and its batts come off with it and the leads unplug at the compartment, so the amplifier, the L-pad and the leads can be reached.
7. Four levelling feet (M8, 40 swivel bases on rubber pads) in inserts in the bottom, 50 in from its edges, lift the cabinet 12 mm: set them so it stands level and cannot rock under the horn, and lock their nuts.
8. No wood: every panel is cast acrylic, 20 mm, tinted through in its colour (red, orange, amber, green, blue, violet, frosted, black), recycled (made from recycled MMA) where a maker can make the colour, cut on a CNC router from 4 × 8 ft sheets of one batch (approve a tint swatch and a cemented corner sample first). The ring (6 mm) is cast acrylic tinted opaque in the ring colour; the cloth disc's frame (12 mm) black cast acrylic.

Notes

barkhorn: Two-way horn loudspeaker after Friendly Pressure, the Cloud: tinted cast acrylic, no wood

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Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

9. Assembly: each panel's edges milled square and polished where they show, the panels annealed (about 80 C, an hour for each 3 mm of thickness, cooled slowly); the ribs cemented to their panels flat (the back's to the back); the shell's corners, the front panel (set back 24 mm), the two braces and the back's four land strips cemented with a two-part polymerising acrylic cement for thick sheet (ACRIFIX 2R 0190 or Weld-On 40), clamped square on a jig, no visible bubble in any joint; the box annealed again; the lands' faces, the bosses' and the seal's groove milled in one setup; the back closes the box last, seated hard on the lands and the braces' bosses over a cord seal, twenty-two screws, and comes off. No solvent cement: it crazes thick cast sheet.
10. The woofer flush in its rebate on a gasket, on eight M6 brass inserts bonded with epoxy in 8.5 holes; the ring over its frame on its magnets, bonded in their pockets; each port in the cabinet's own acrylic, its tube cemented into its 8 mm frame, fitted from the front into a recess on a foam gasket, four black M4 screws in its corners into tapped threads; the tubes cut to the length found on the first sample before their ends are flared.
11. The front: the panel's own tinted acrylic, the lining showing through it round the woofer, and a cloth disc over the woofer in the acrylic's colour, its frame standing on the ring on four magnet pairs.
12. Seen through: everything inside is drawn to be seen. The leads run straight and tied, the glands and the brass sockets show, the amplifier's compartment is black anodised, and each batt's voile sleeve has its seams to the wall.
13. Lifting (round 24): a grip 240 × 42 for both hands through each side, 92 below the top, a cup of 3 mm acrylic cemented behind it and the side's batt cut round it; two people lift a speaker by its grips, the horn off. The feet's M8 inserts take castors, spikes or isolation feet in place of the levelling feet, sold separately.
14. The mark (round 24): EARF 140 wide, laser-etched high on the back, centred, the etch filled with the speaker's own colour, deeper.

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